

Mining Guest Experience from Airbnb Reviews: Evidence from Barcelona Airbnb Listings

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BSE, DSM, DSDM, Cooking Metadata

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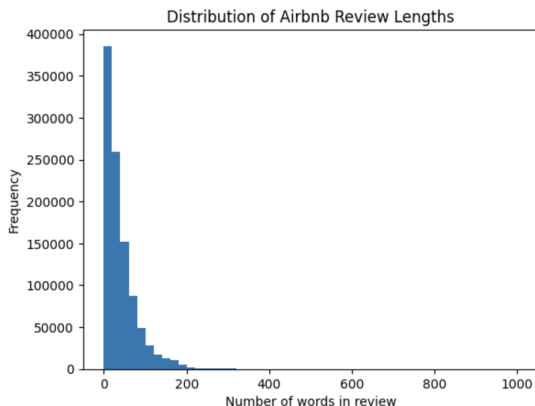
Agenda

- 1 Introduction
- 2 Methodology
- 3 EDA
- 4 TF-IDF
- 5 LDA
- 6 Supervised Classification
- 7 Limitations

- **Research Question:** Beyond the purchase price and rental income, what property characteristics are associated with positive experiences, leading potentially to higher rates of renting?
- **NLP Perspective:** We combine a real estate perspective with text mining. We use Airbnb review data from Barcelona to understand what guests say in their reviews, which topics appear most frequently, and which words are associated with positive or negative experiences.
- **Project goal:** Use Airbnb reviews to identify what drives positive and negative experiences and advise potential buyers or Airbnb property owner to make better decisions about where to buy and how to improve in terms of service.

- EDA(exploratory data analysis)
- TF-IDF to identify informative words
- More advanced NLP methods: LDA/Supervised Classification

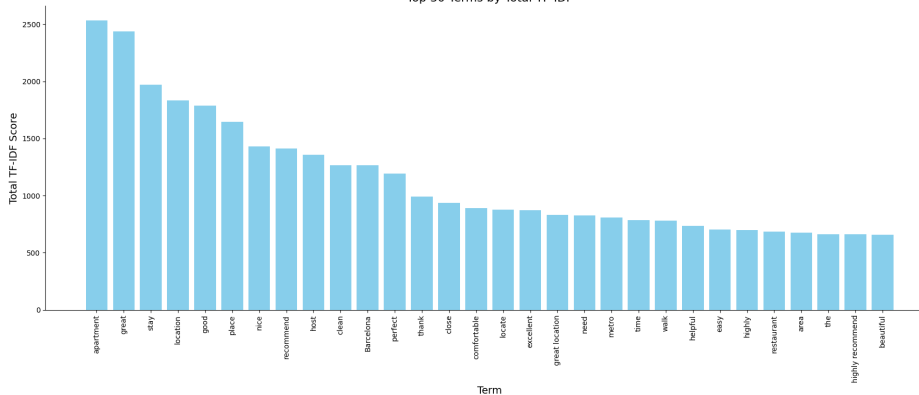
- <https://insideairbnb.com/get-the-data/>
- The first step is translation into English.
- The second step is to delete all unnecessary columns.
- Then we clean the text through standard preprocessing: lowercasing, removing punctuation, removing stopwords, and preparing a clean corpus for analysis.



- **Number of reviews:** 1019270; **Mean of the lengths:** 40
- Most Airbnb reviews are short.
- Shorter reviews may contain less context which not helpful to identify themes and sentiments for LDA.

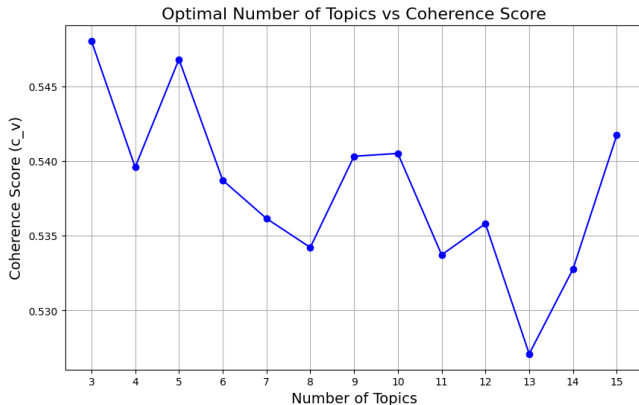
TF-IDF

Top 30 Terms by Total TF-IDF



- First, it helps us identify the most informative terms in the reviews
- Second, it provides numerical text features that we use later in the supervised model.
- Highlights the most informative words in reviews.

LDA Introduction



- We applied a coherence score to get the optimal number of topics.
- The selected model (5 topics) was retrained and interpreted using top words per topic and topic word clouds.

WordCloud



- Related to the distribution of the words, we can see some of them appear in all topics because in general we have reviews with less than 100 words.

- Accessibility and neighborhood convenience are central to how guests evaluate Airbnb listings in Barcelona.
- Operational improvements in communication and cleanliness can influence guest satisfaction.
- The five-topic structure indicates that high satisfaction is multi-factorial: good location, clean and comfortable spaces, and responsive hosting consistently appear as the main drivers of favorable reviews.

Supervised Learning Pipeline

- **Label Construction:** Generate sentiment labels using polarity scores
- **Train-Test Split:** Divide data into training and testing sets
- **Model Training:** Fit Logistic Regression model
- **Prediction:** Classify reviews as positive or negative
- **Evaluation:** Assess performance using accuracy and confusion matrix
- **Interpretation:** Analyze model coefficients to identify key drivers

Model Evaluation: Confusion Matrix

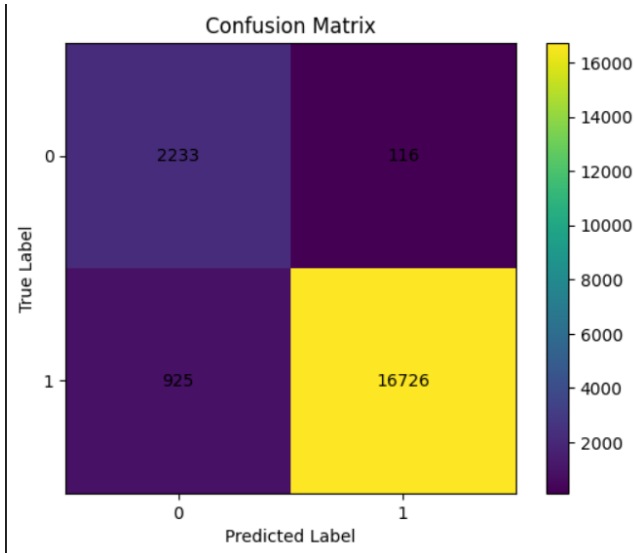


Figure: Confusion matrix of the logistic regression classifier

Most predictive words for Review Sentiment

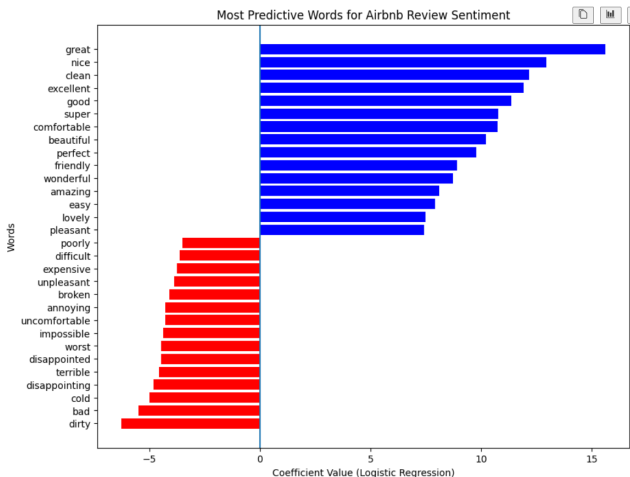


Figure: Most predictive words with their associated coefficient

Topic Sentiment Impact

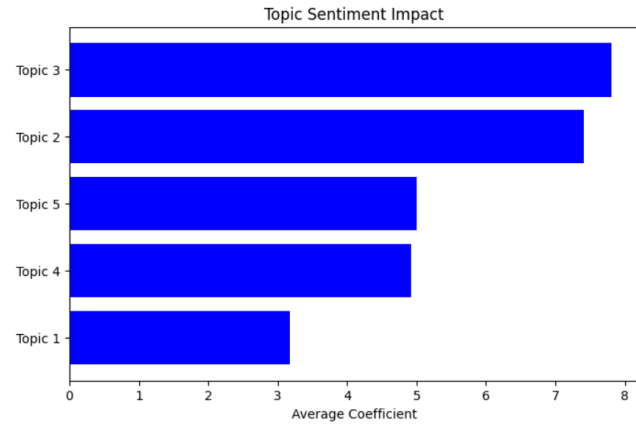


Figure: Topic Sentiment Impact for each topic

- The review corpus is heavily dominated by positive and neutral language, so negative experiences are underrepresented.
- These patterns affect not only TF-IDF, but also later NLP methods such as LDA, where topics may reflect general praise instead of property-specific features.

Thank you!
Questions?